

❖ Ch2_No.17

If the radius of the beam waist of the Gaussian beam is $w_0=0.3$ mm and the wavelength is $\lambda=632.8$ nm. Calculate the q parameter at the waist, 30 cm away from the beam waist, and the infinite distance from the beam waist, respectively.

$$w_0 = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}$$

$$\lambda = 632.8 \text{ nm} = 632.8 \times 10^{-9} \text{ m}$$

⇒ Rayleigh:

$$z_R = \frac{\pi w_0^2}{\lambda} = \frac{\pi \cdot (3 \times 10^{-4} \text{ m})^2}{632.8 \times 10^{-9} \text{ m}} = 44.68 \text{ cm}$$

$$\Rightarrow q(z) = z + i \cdot z_R$$

$$z=0 : q(0) = 0 + i \times 44.68 \text{ cm} = \underline{44.68 i \text{ cm}}$$

$$z=30 \text{ cm} : q(30) = \underline{30 + 44.68 i \text{ cm}}$$

$$z \rightarrow \infty : q(\infty) = \infty + 44.68 i \text{ cm} \approx \underline{\infty}$$

❖ Ch2_No.22

The wavelength of the CO_2 laser is $\lambda=10.6$ μm , and the radius of the beam waist is $w_0=3$ mm. If we use a convex lens with $F=2$ cm to focus the laser beam, then where the position of lens should be if the radius of the focused beam is $w_0'=20$ μm and 2.5 μm .

$$\lambda = 10.6 \mu\text{m}$$

$$w_0 = 3 \text{ mm}$$

$$F = 2 \text{ cm}$$

$$\Rightarrow z_R = \frac{\pi w_0^2}{\lambda} = \frac{\pi \cdot (3 \text{ mm})^2}{10.6 \mu\text{m}} \approx 266.71 \text{ cm}$$

$$\Rightarrow M = \frac{w_0'}{w_0}$$

$$\Rightarrow \frac{1}{M^2} = \left(1 - \frac{z}{F}\right)^2 + \left(\frac{z_R}{F}\right)^2$$

$$\textcircled{1} \quad w_0 = 20 \mu\text{m}$$

$$\Rightarrow M = \frac{20 \mu\text{m}}{3 \text{ mm}} = \frac{1}{150}$$

$$\checkmark \left(\frac{z_R}{F}\right)^2 = \left(\frac{266.71}{2}\right)^2 \approx 17784.1$$

$$\Rightarrow 150^2 = \left(1 - \frac{z}{2}\right)^2 + \left(\frac{z_R}{F}\right)^2$$

$$\left(1 - \frac{z}{2}\right)^2 = 4715.9$$

$$\Rightarrow 1 - \frac{z}{2} = \pm 68.67$$

$$\Rightarrow z_1 \approx \underline{-135.34 \text{ cm}} \quad \text{front}$$

$$z_2 \approx \underline{139.34 \text{ cm}}$$

behind $\curvearrowright$ In real engineering prefer +

$$\textcircled{2} \quad w_0' = 2.5 \mu\text{m}$$

$$M = \frac{2.5 \mu\text{m}}{3 \text{ mm}} = \frac{1}{1200}$$

$$\Rightarrow (1200)^2 = \left(1 - \frac{z}{2}\right)^2 + \left(\frac{z_R}{F}\right)^2$$

$$\left(1 - \frac{z}{2}\right)^2 = 1422215.9$$

$$\Rightarrow 1 - \frac{z}{2} = \pm 1192.57$$

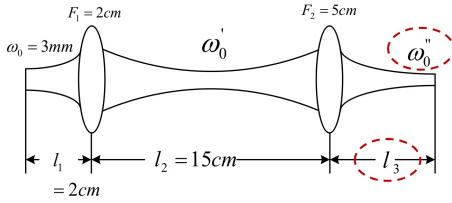
$$\Rightarrow z_1 \approx \underline{-2383.14 \text{ cm}} \quad \text{front}$$

$$z_2 \approx \underline{2387.14 \text{ cm}}$$

behind $\curvearrowright$ In real engineering prefer +

❖ Ch2_No.23

As shown in the figure below, the wavelength of the incident light beam is $\lambda=10.6 \mu\text{m}$. Calculate the value of w_0'' and l_3 .



$$\lambda = 10.6 \mu\text{m} \quad , \quad w_0 = 3 \mu\text{m}$$

$$F_1 = 2 \text{ cm} \quad , \quad z_1 = l_1 = 2 \text{ cm}$$

$$l_2 = 15 \text{ cm}$$

$$F_2 = 5 \text{ cm}$$

$$\Rightarrow l_1 = F_1 = 2 \text{ cm} \quad \hookrightarrow \quad l_1' = 2 \text{ cm}$$

$$w_0' = \frac{\lambda F_1}{\pi w_0} = \frac{10.6 \mu\text{m} \cdot 2 \text{ cm}}{\pi \cdot 3 \text{ mm}} \approx 22.49 \mu\text{m}$$

$$\Rightarrow z_{R2} = \frac{\pi (w_0')^2}{\lambda} = \frac{\pi (22.49 \mu\text{m})^2}{10.6 \mu\text{m}} \approx 0.15 \text{ mm}$$

$$z_2 = l_2 - l_1' = 13 \text{ cm}$$

$$\begin{aligned} \Rightarrow l_3 &= F_2 + \frac{(z_2 - F_2) \cdot F_2^2}{(z_2 - F_2)^2 + z_{R2}^2} \\ &= 5 \text{ cm} + \frac{(13 \text{ cm} - 5 \text{ cm}) 5 \text{ cm}^2}{(13 \text{ cm} - 5 \text{ cm})^2 + (0.15 \text{ mm})^2} \end{aligned}$$

$$\approx \underline{8.125 \text{ cm}}$$

$$\Rightarrow w_0'' = \frac{w_0'}{\sqrt{\left(1 - \frac{z_2}{F_2}\right)^2 + \left(\frac{z_{R2}}{F_2}\right)^2}} \approx \underline{14.06 \mu\text{m}}$$